

Exploratory Data Analysis

STOR 320.002 Group 16

April 27, 2026

Load Necessary Libraries

```
suppressPackageStartupMessages(library(tidyverse))

## Warning: package 'purrr' was built under R version 4.5.3

suppressPackageStartupMessages(library(janitor))
suppressPackageStartupMessages(library(funModeling))
suppressPackageStartupMessages(library(skimr))
suppressPackageStartupMessages(library(readxl))
suppressPackageStartupMessages(library(readr))
suppressPackageStartupMessages(library(lubridate))
suppressPackageStartupMessages(library(DataExplorer))
suppressPackageStartupMessages(library(dlookr))
suppressPackageStartupMessages(library(knitr))
suppressPackageStartupMessages(library(gt))
suppressPackageStartupMessages(library(ggribes))
suppressPackageStartupMessages(library(PerformanceAnalytics))

## Warning: package 'PerformanceAnalytics' was built under R version 4.5.3

## Warning: package 'xts' was built under R version 4.5.3

knitr::opts_chunk$set(echo = TRUE)
options(scipen = 999) # big number = strongly discourages scientific notation

knitr::opts_chunk$set(
  echo = TRUE,
  tidy = TRUE,
  tidy.opts = list(
    width.cutoff = 80,
    arrow = TRUE, # makes pipes look nicer
    indent = 2
  ),
  message = FALSE,
  warning = FALSE
)

options(width = 90) # helps with printed tables too
```

Statement of the Business Problem.

What is the “Business” Problem?

- Consumers, automakers, and policymakers face a growing challenge in navigating the tension between vehicle design and performance on one hand, and competing pressures on the other, including after-purchase financial realities such as rising fuel costs, alongside fuel efficiency regulations and environmental protection measures. This has made understanding real-world fuel costs increasingly critical for consumers, manufacturers, and regulators alike.

What are the goals that the “business” wants to achieve?

- The goal of this analysis is to explore the 2025 EPA Fuel Economy dataset which contains 868 vehicle configurations across 23 manufacturers and 20 vehicle classes, to identify the key vehicle features that are most associated with annual fuel cost. By doing so, we aim to clarify which design and technology choices drive the cost of fueling a vehicle, and where the biggest gaps between vehicles lie.

How does the business currently work?

- The EPA currently tests every new vehicle model annually and publishes city, highway, and combined MPG estimates, GHG ratings on a 1-10 scale, and annual fuel cost estimates through the Fuel Economy Guide and [fueleconomy.gov](https://www.fueleconomy.gov). However, these figures tell consumers what a vehicle’s predicted fuel cost is, but not what the underlying factors are that cause these levels of costs, leaving the relationship between specific design features and fuel cost outcomes largely opaque to both consumers and analysts.

In what ways could a predictive analytics model help to address the business problem?

- A predictive model built on this data could estimate a vehicle’s annual fuel cost from its design features and fuel efficiency characteristics, such as engine displacement, drive type, and fuel efficiency. This would help consumers anticipate the true long-term cost of owning a vehicle beyond the sticker price, help automakers identify which feature combinations minimize fuel costs for consumers, and help policymakers evaluate whether current regulations are effectively incentivizing designs that reduce the financial burden of fuel on drivers, ultimately supporting better decisions across all three groups.

Clean the Data and Take A Glimpse

```
knitr::opts_chunk$set(echo = TRUE)
library(tidyverse)

# Import Data Below
# read_csv()

cars_fe <- read_csv("Group 16 2025 Fuel Economy Dataset(FEguide).csv") |>
  clean_names() |>
  mutate(across(where(is.character), ~ str_squish(.))) |>
  mutate(across(where(is.character),
    ~ na_if(.x, ""))) |>
  mutate(across(where(is.character),
    ~ case_when(
      .x %in% c("", " ", "NA", "NULL", "N/A", "?", "-") ~
NA_character_,
      TRUE ~ .x)))
```

Convert all Characters to Factors

```
cars_fe <- cars_fe |>
  mutate(
    across(where(is.character), as.factor),
    release_date = as.Date(release_date, format = "%m/%d/%Y")
  )
```

Extracting all Numeric Features

```
cars_fe_num_data <- cars_fe |>
  select_if(is.numeric)
```

Skimming all Numeric Features

```
num_s <- skim_without_charts(cars_fe_num_data)
```

Extracting Some Interesting Outputs

```
focus_table <- as.data.frame(num_s) |>
  select(
    skim_variable,
    numeric.p0,
    numeric.p25,
    numeric.mean,
    numeric.p50,
    numeric.p75,
    numeric.p100,
    numeric.sd
  )
```

Adding a Count Column to the Table

```
focus_table$count <- rep(868, 15)
focus_table <- focus_table |>
  relocate(count, .before = 2)
```

Constructing Numeric Data Quality Report

```
extra_two <- funModeling::status(cars_fe_num_data)[, c("p_na", "unique")]
rownames(extra_two) <- NULL

numeric_report_table <- cbind(focus_table, extra_two)
numeric_report_table <- numeric_report_table |>
  relocate(p_na, .before = 3) |>
  relocate(unique, .before = 4)

names(numeric_report_table) <- c("Feature",
                                "Count",
                                "% Missing",
                                "Cardinality",
                                "Min",
                                "1st Qrt.",
                                "Mean",
                                "Median",
                                "3rd Qrt.",
                                "Max",
                                "Std. Dev.")
```

Storing Categorical Features

```
cars_fe_cat_data <- cars_fe |>
  select_if(is.factor)
```

Skimming and Viewing Data Summary of Categorical Features as a Whole

```
cars_fe_cat_data |>
  skim() |>
  summary()
```

Data summary

Name	cars_fe_cat_data
Number of rows	868
Number of columns	12

Column type frequency:

factor	12
--------	----

Group variables

None

Storing Detailed Statistics for Every Individual Variable

```
cat_s <- skim_without_charts(cars_fe_cat_data)
```

Creating a Focus Table

```
focus_table_2 <- as.data.frame(cat_s) |>
  select(
    skim_variable,
    n_missing,
    complete_rate,
    factor.n_unique,
    factor.top_counts
  )
```

Constructing Categorical Data Quality Report

```
focus_table_2 <- focus_table_2 |>
  mutate(Count = nrow(cars_fe))

cat_report_table <- focus_table_2 |>
  tidyr::extract(
    factor.top_counts,
    into = c("Mode", "Mode Freq.", "2nd Mode", "2nd Mode Freq."),
    regex = "(.+?):\\s*(\\d+), \\s*(.+?):\\s*(\\d+)",
    remove = TRUE
  ) |>

  mutate(
    `Mode Freq.` = as.numeric(`Mode Freq.`),
    `2nd Mode Freq.` = as.numeric(`2nd Mode Freq.`),
    `% Miss.` = round(100 * n_missing / Count, 2),
    `Mode %` = round(100 * `Mode Freq.` / Count, 2),
    `2nd Mode %` = round(100 * `2nd Mode Freq.` / Count, 2)
  ) |>

  rename(
    Feature = skim_variable,
    Cardinality = factor.n_unique
  ) |>

  select(
    Feature,
    Count,
    `% Miss.` ,
    Cardinality,
```

```
Mode,  
`Mode Freq.` ,  
`Mode %` ,  
`2nd Mode` ,  
`2nd Mode Freq.` ,  
`2nd Mode %`  
)
```

Final Table Data Quality Report: Numeric Features

numeric_report_table						
##	Feature	Count	% Missing	Cardinality	Min	1st Qrt.
##	Mean					
## 1	model_year	868	0.00000000	1	2025.0000	2025.00000
2025.000000						
## 2	engine_displacement	868	0.00000000	32	1.2000	2.00000
2.956567						
## 3	num_of_cylinders	868	0.00000000	7	3.0000	4.00000
5.297235						
## 4	city_fe	868	0.00000000	45	8.0000	17.00000
21.754608						
## 5	hwy_fe	868	0.00000000	41	12.0000	23.00000
27.428571						
## 6	comb_fe	868	0.00000000	44	9.0000	19.00000
23.892857						
## 7	city_unadj_fe	868	0.00000000	555	9.5000	21.32960
28.154119						
## 8	hwy_unadj_fe	868	0.00000000	581	16.5000	32.49907
39.559594						
## 9	comb_unadj_fe	868	0.00000000	738	11.7416	25.24697
32.153329						
## 10	num_gears	868	0.00000000	7	1.0000	7.00000
7.668203						
## 11	max_ethanol_percent	868	0.02073733	3	10.0000	10.00000
17.694118						
## 12	max_biodiesel_percent	868	0.97926267	1	20.0000	20.00000
20.000000						
## 13	annual_fuel_cost	868	0.00000000	53	900.0000	2100.00000
2641.647465						
## 14	ghg_rating	868	0.00000000	8	1.0000	4.00000
4.691244						
## 15	x4dr_pass_vol	868	0.75576037	32	87.0000	94.00000
100.259434						
##	Median	3rd Qrt.	Max	Std. Dev.		
## 1	2025.00000	2025.00000	2025.0000	0.000000		
## 2	2.70000	3.50000	8.0000	1.170944		
## 3	4.00000	6.00000	16.0000	1.714755		
## 4	20.00000	25.00000	57.0000	7.422085		
## 5	26.00000	32.00000	58.0000	6.698704		
## 6	23.00000	27.00000	57.0000	6.995882		
## 7	25.50440	31.80000	82.8000	10.873110		
## 8	37.79105	46.12728	77.8000	9.741842		
## 9	29.88570	36.98208	80.4727	10.206697		
## 10	8.00000	9.00000	10.0000	2.153949		
## 11	10.00000	15.00000	85.0000	19.838288		
## 12	20.00000	20.00000	20.0000	0.000000		
## 13	2550.00000	3100.00000	7100.0000	783.904509		

## 14	5.00000	5.00000	8.0000	1.299123
## 15	99.00000	102.00000	157.0000	11.504940

Final Table Data Quality Report: Categorical Features

cat_report_table

##		Feature	Count	% Miss.	Cardinality	Mode	Mode	Freq.	Mode
##	% 2nd Mode								
## 1		mfr_name	868	0.00	23	Gen		144	
16.59	Toy								
## 2		division	868	0.00	41	TOY		78	
8.99	BMW								
## 3		carline	868	0.00	670	SIL		9	
1.04	SIL								
## 4		trans_description	868	0.00	7	Sem		382	
44.01	Aut								
## 5		lockup_torque_converter	868	0.00	2	Y		673	
77.53	N								
## 6		trans_creeper_gear	868	0.00	2	N		860	
99.08	Y								
## 7		drive_desc	868	0.00	5	All		353	
40.67	2-W								
## 8		fuel_usage_desc	868	0.00	5	Gas		415	
47.81	Gas								
## 9		descriptor_model_type	868	7.49	11	SID		402	
46.31	SID								
## 10		carline_class_desc	868	0.00	20	Sma		170	
19.59	Sta								
## 11		stop_start_system_status	868	0.00	2	Yes		708	
81.57	No								
## 12		air_aspiration	868	0.00	5	Tur		557	
64.17	Nat								
##	2nd Mode	Freq.	2nd Mode	%					
## 1		115		13.25					
## 2		73		8.41					
## 3		9		1.04					
## 4		211		24.31					
## 5		195		22.47					
## 6		8		0.92					
## 7		169		19.47					
## 8		225		25.92					
## 9		172		19.82					
## 10		134		15.44					
## 11		160		18.43					
## 12		266		30.65					

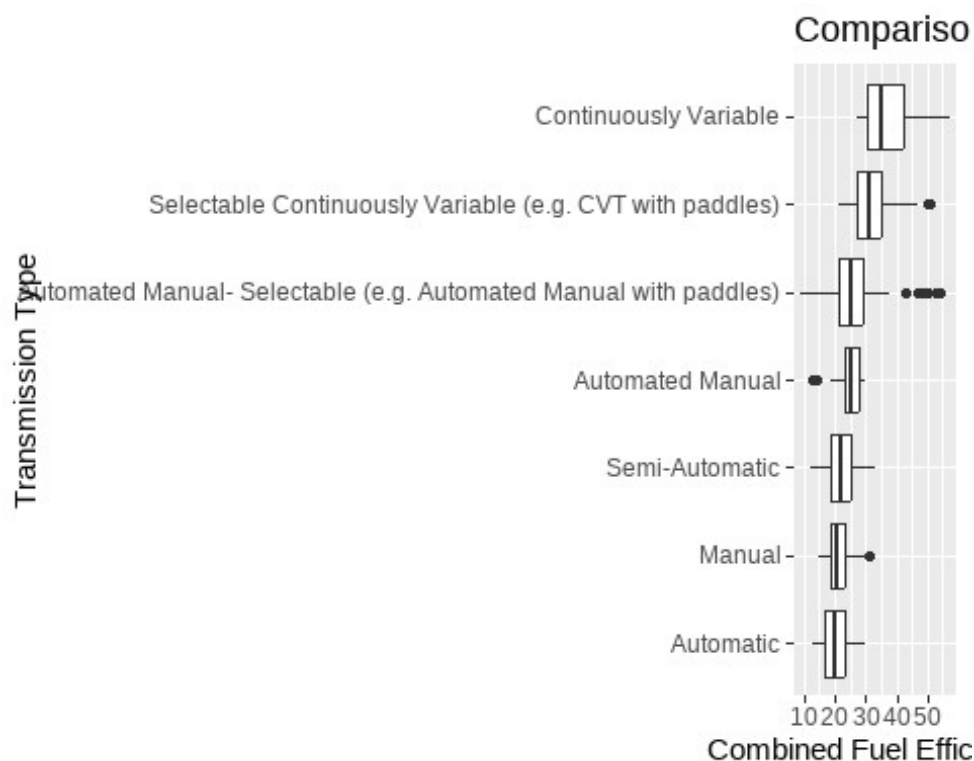
Data Issues and Outline of proposed Handling Techniques.

- In our data, the features `x4dr_pass_vol` and `max_biodiesel_percent` are missing over 50% of their values. This is because these categories do not apply to every car. The maximum biodiesel percent has 97.93% missing with a cardinality of 1. We have decided to handle this by removing the variable and not analyzing it going forward. The feature `x4dr_pass_vol` or the passenger volume in a 4-door car has a missing percentage of 75.57%. While this percentage is very high, it is a variable we have been exploring; it will end up being only applicable to 4-door vehicles. We can also remove the `model_year` variable because our dataset is strictly 2025, so all the vehicles have the same model year.

Creator: Annabelle Lawson

Q1: Is there an association between the transmission type and the combined (guide) FE?

```
ggplot(cars_fe, aes(x = comb_fe, y = fct_reorder(trans_description, comb_fe, median))) +  
  geom_boxplot() +  
  labs(  
    title = "Comparison of Transmission Types using Combined Fuel  
Efficiency",  
    x = "Combined Fuel Efficiency",  
    y = "Transmission Type"  
  )
```

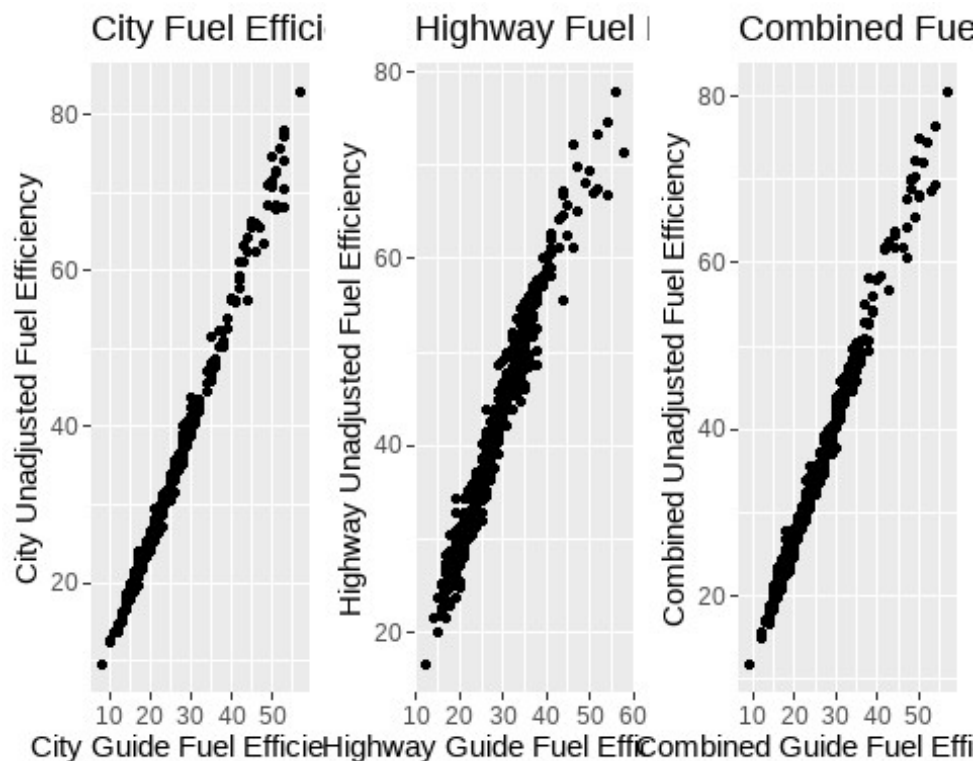


Q2: What is the correlation between the guide and unadjusted FE between city, hwy, and combined?

```
library(cowplot)  
city <- ggplot(cars_fe, aes(x = city_fe, y = city_unadj_fe)) +  
  geom_point() +  
  labs(  
    title = "City Fuel Efficiency: Guide vs Unadjusted",  
    x = "City Guide Fuel Efficiency",  
    y = "City Unadjusted Fuel Efficiency"  
  )
```

```
hwy <- ggplot(cars_fe, aes(x = hwy_fe, y = hwy_unadj_fe))+
  geom_point()+
  labs(
    title = "Highway Fuel Efficiency:Guide vs Unadjusted",
    x = "Highway Guide Fuel Efficiency",
    y = "Highway Unadjusted Fuel Efficiency"
  )

comb <- ggplot(cars_fe, aes(x = comb_fe, y = comb_unadj_fe))+
  geom_point()+
  labs(
    title = "Combined Fuel Efficiency:Guide vs Unadjusted",
    x = "Combined Guide Fuel Efficiency",
    y = "Combined Unadjusted Fuel Efficiency"
  )
plot_grid(city, hwy, comb, ncol = 3)
```



Interpreter: Cooper Lair

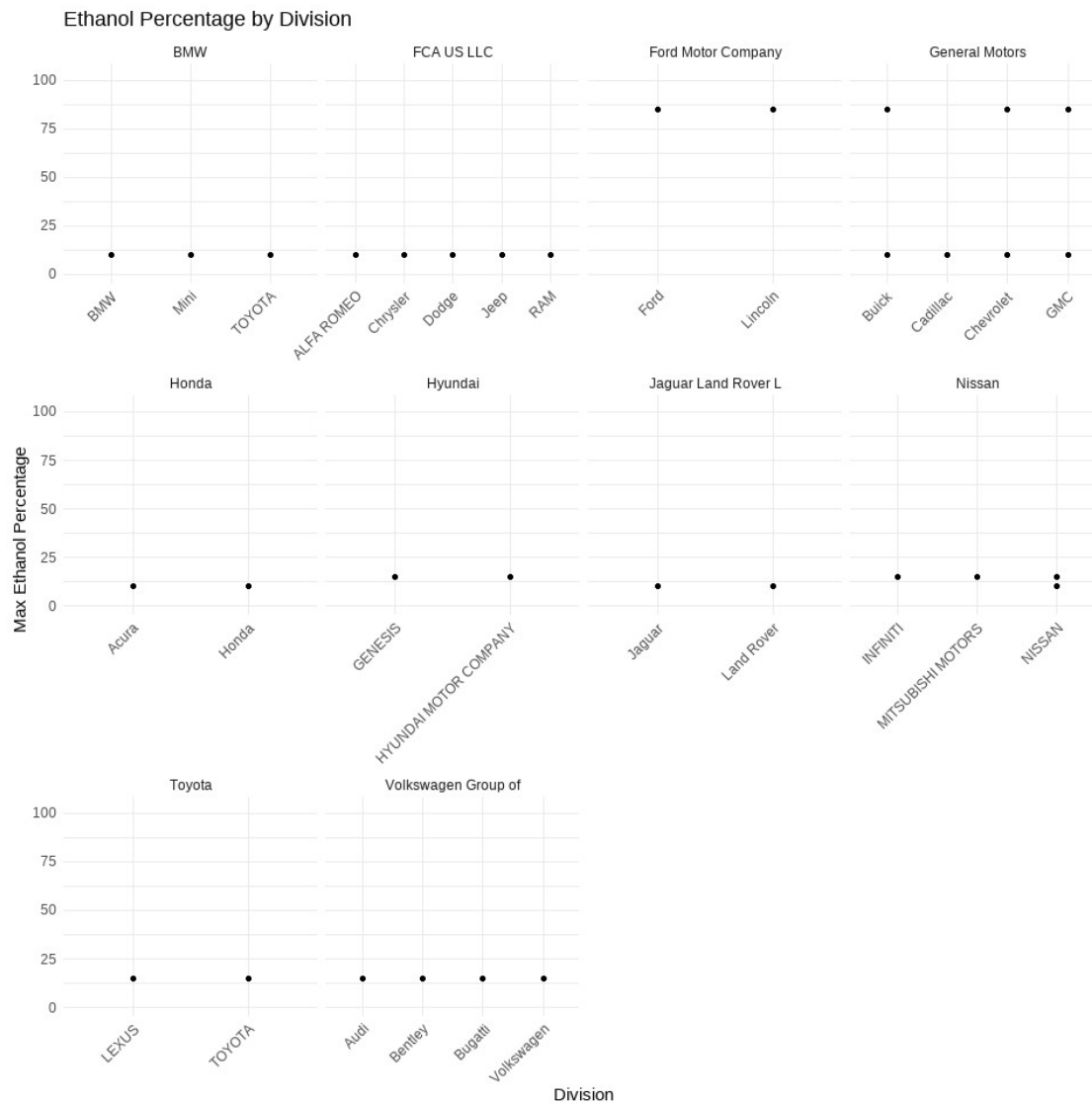
Q1: Are there ethanol percentage differences across different divisions of the same manufacturer?

```
multi_div_mfr <- cars_fe |>
  group_by(mfr_name) |>
```

```

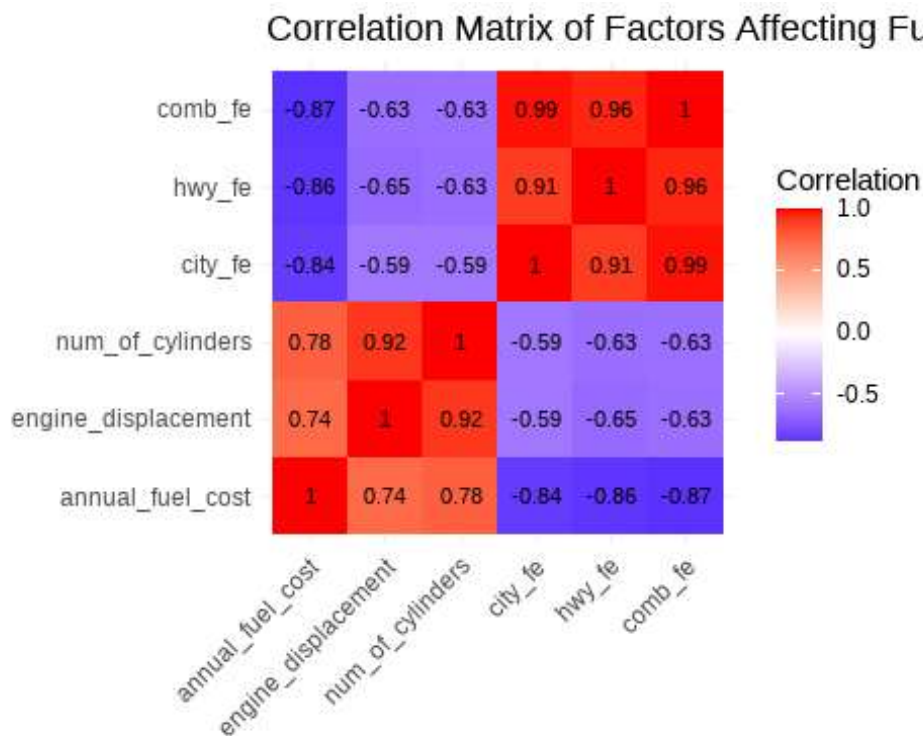
filter(n_distinct(division) > 1) |>
ungroup()
multi_div_mfr |>
filter(!is.na(max_ethanol_percent)) |>
ggplot(aes(x = division, y = max_ethanol_percent)) +
  geom_point() +
  facet_wrap(~ mfr_name, scales = "free_x") +
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.2))) +
  coord_cartesian(ylim = c(0, 90)) +
  labs(
    title = "Ethanol Percentage by Division",
    x = "Division",
    y = "Max Ethanol Percentage"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



Q2: Are there a set of factors that can predict the average cost of fuel?

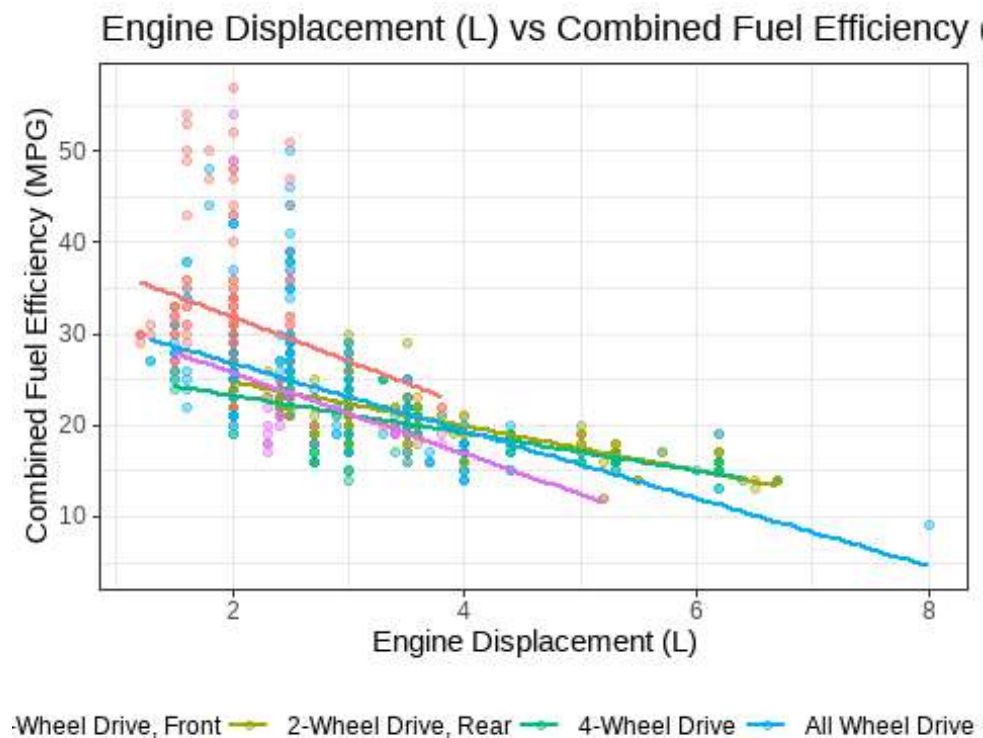
```
num_data <- cars_fe_num_data |>
  select(
    annual_fuel_cost,
    engine_displacement,
    num_of_cylinders,
    city_fe,
    hwy_fe,
    comb_fe
  ) |>
  na.omit()
cor_matrix <- cor(num_data)
cor_df <- as.data.frame(as.table(cor_matrix))
ggplot(cor_df, aes(Var1, Var2, fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = round(Freq, 2)), size = 3) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint =
0) +
  labs(
    title = "Correlation Matrix of Factors Affecting Fuel Cost",
    x = "",
    y = "",
    fill = "Correlation"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Orator 1: SAMAY SURATWALA

Q1: How does the correlation between engine displacement and combined fuel efficiency vary when comparing different drive types (AWD vs FWD vs RWD)?

```
ggplot(cars_fe,
       aes(x = engine_displacement, y = comb_fe, color = drive_desc)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Engine Displacement (L) vs Combined Fuel Efficiency (MPG) with Drive Type",
    x = "Engine Displacement (L)",
    y = "Combined Fuel Efficiency (MPG)",
    color = "Drive Type"
  ) +
  theme_bw() +
  theme(legend.position = "bottom")
```



Q2: Is there a significant Greenhouse gas gap between cars with 4-cylinder engines vs 6-cylinder engines, or has modern turbocharging narrowed that gap?

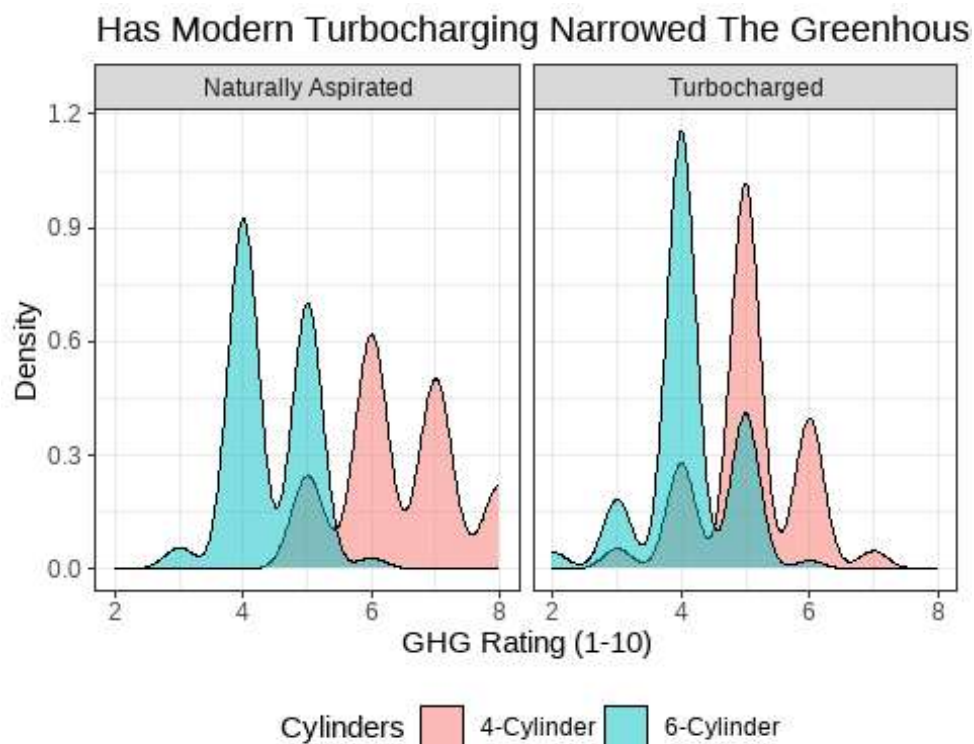
#

```

cyl_ghg_asp <- cars_fe |>
  filter(num_of_cylinders %in% c(4, 6)) |>
  mutate(
    chosen_cylinders = factor(num_of_cylinders, labels = c("4-Cylinder", "6-
Cylinder")),
    air_aspiration = factor(air_aspiration)
  ) |>
  filter(air_aspiration %in% c("Naturally Aspirated", "Turbocharged")) |>
  mutate(air_aspiration = droplevels(air_aspiration))

ggplot(cyl_ghg_asp, aes(x = ghg_rating, fill = chosen_cylinders)) +
  geom_density(alpha = 0.5) +
  facet_wrap(~air_aspiration, ncol = 2) +
  labs(
    title = "Has Modern Turbocharging Narrowed The Greenhouse Gas (GHG)
Gap?",
    x = "GHG Rating (1-10)",
    y = "Density",
    fill = "Cylinders"
  ) +
  theme_bw() +
  theme(legend.position = "bottom")

```



```

ggplot(cyl_ghg_asp, aes(x = air_aspiration,
  y = ghg_rating,

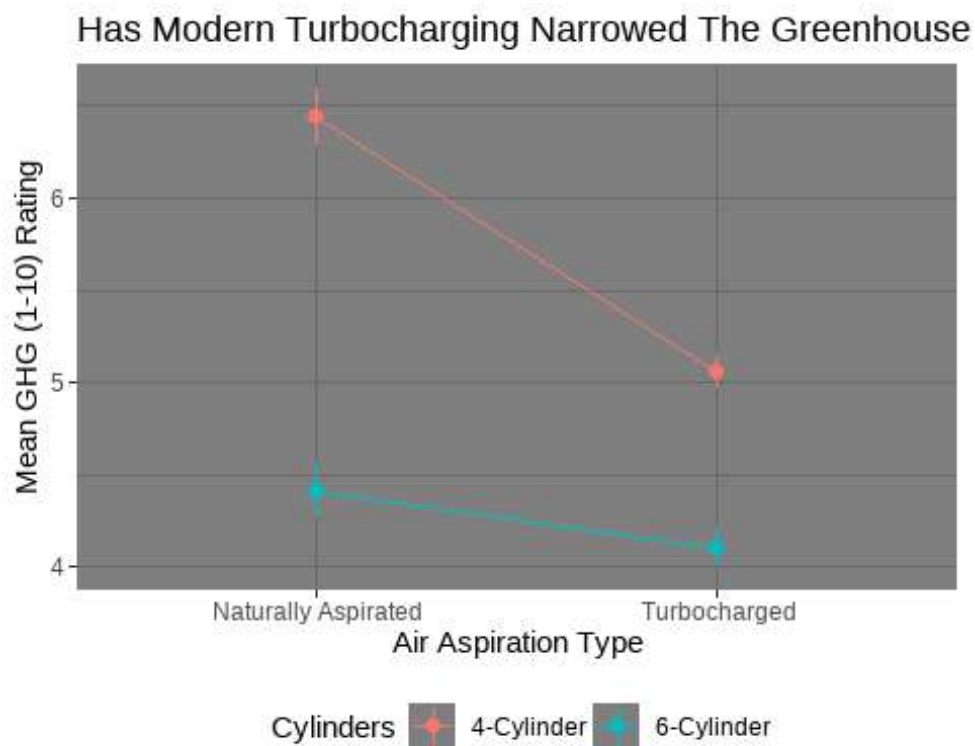
```



```

        color = chosen_cylinders,
        group = chosen_cylinders)) +
stat_summary(fun.data = mean_cl_normal, geom = "pointrange") +
geom_line(stat = "summary", fun = mean) +
labs(
  title = "Has Modern Turbocharging Narrowed The Greenhouse Gas (GHG)
Gap?",
  x = "Air Aspiration Type",
  y = "Mean GHG (1-10) Rating",
  color = "Cylinders"
) +
theme_dark() +
theme(legend.position = "bottom")

```



Deliverer: Akshita Padala

Q1: Does drive type (AWD vs FWD vs RWD) have a different efficiency penalty on vehicles of different car-line class descriptions (e.g., “Small SUV”, “Two Seaters”)?

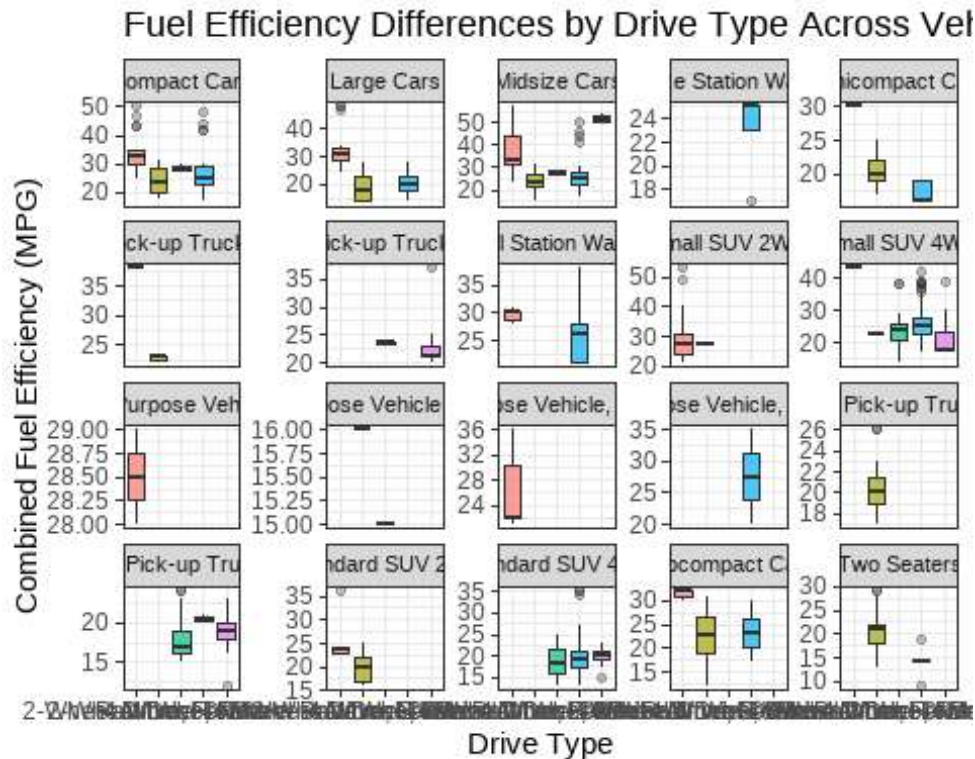
```
cars_fe |>
  filter(!is.na(carline_class_desc), !is.na(drive_desc), !is.na(comb_fe)) |>
  mutate(
    drive_desc = droplevels(drive_desc),
    carline_class_desc = droplevels(carline_class_desc)
  ) |>

  ggplot(aes(x = drive_desc, y = comb_fe, fill = drive_desc)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.3) +

  facet_wrap(~ carline_class_desc, scales = "free_y") +

  labs(
    title = "Fuel Efficiency Differences by Drive Type Across Vehicle
Classes",
    x = "Drive Type",
    y = "Combined Fuel Efficiency (MPG)",
    fill = "Drive Type"
  ) +

  theme_bw() +
  theme(
    legend.position = "none",
    strip.text = element_text(size = 8)
  )
```



Q2: Is there a relationship between number of gears in a transmission and combined fuel efficiency?

```
cars_fe |>
  filter(!is.na(num_gears), !is.na(comb_fe)) |>

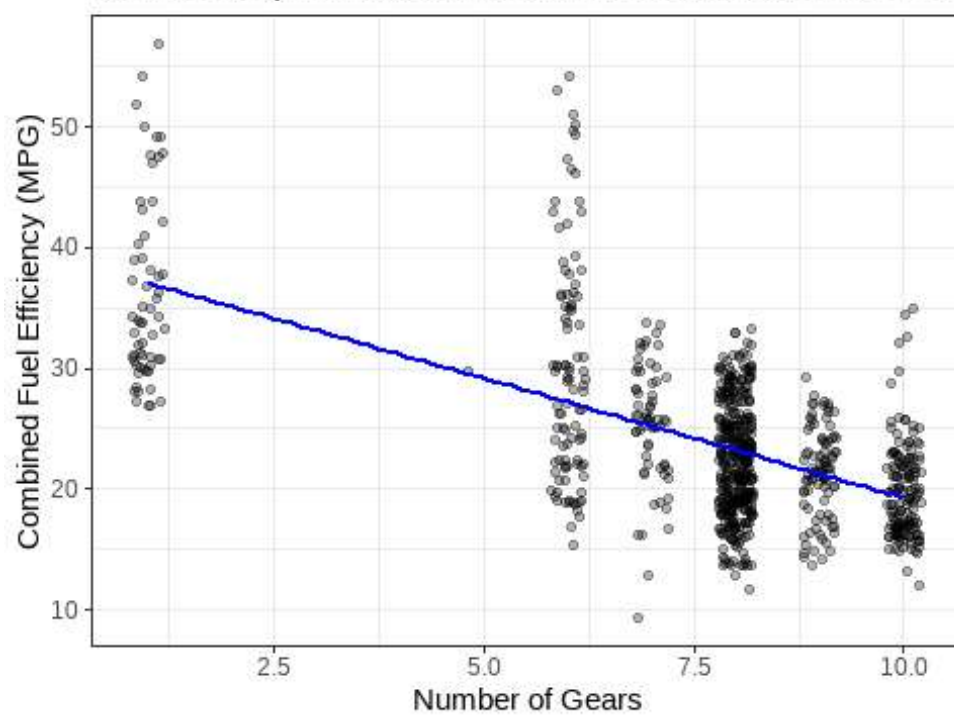
  ggplot(aes(x = num_gears, y = comb_fe)) +
  geom_jitter(alpha = 0.3, width = 0.2) +

  geom_smooth(method = "lm", se = FALSE, color = "blue") +

  labs(
    title = "Relationship Between Number of Gears and Fuel Efficiency",
    x = "Number of Gears",
    y = "Combined Fuel Efficiency (MPG)"
  ) +

  theme_bw()
```

Relationship Between Number of Gears and Fuel Efficiency

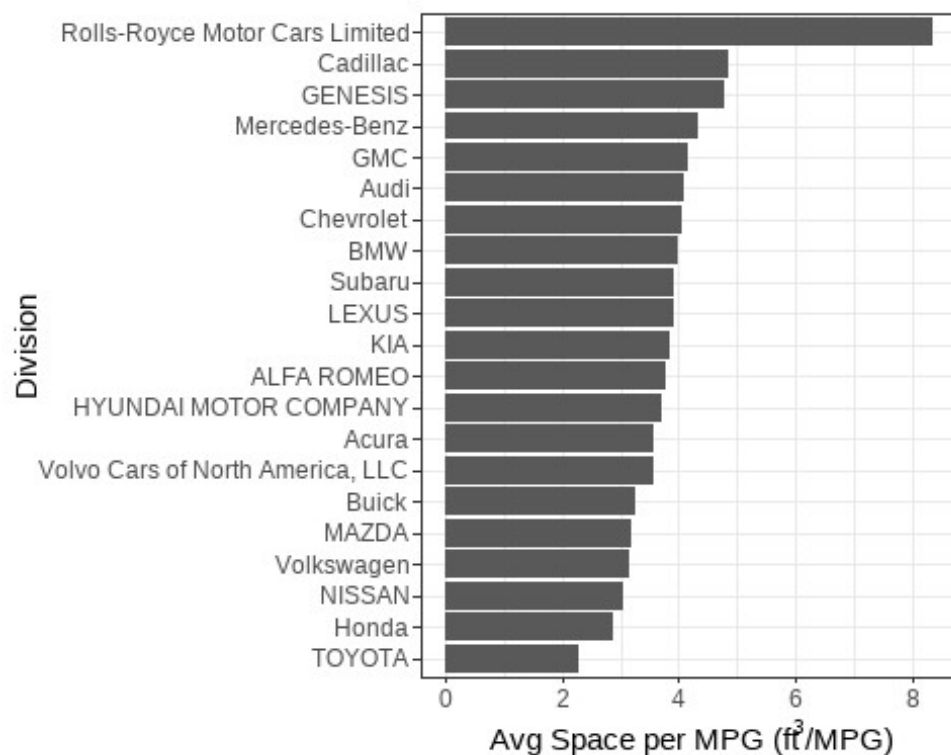


Orator 2: Alexa Kisłowski

Q1: Which car manufacturers are best at efficient packaging - providing the most interior passenger space per gallon of fuel consumed?

```
efficiency <- cars_fe |>
  mutate(space_per_mpg = x4dr_pass_vol / comb_fe) |>
  group_by(division) |>
  summarise(
    avg_space_per_mpg = mean(space_per_mpg, na.rm=TRUE)) |>
  arrange(avg_space_per_mpg) |>
  filter(!is.nan(avg_space_per_mpg))

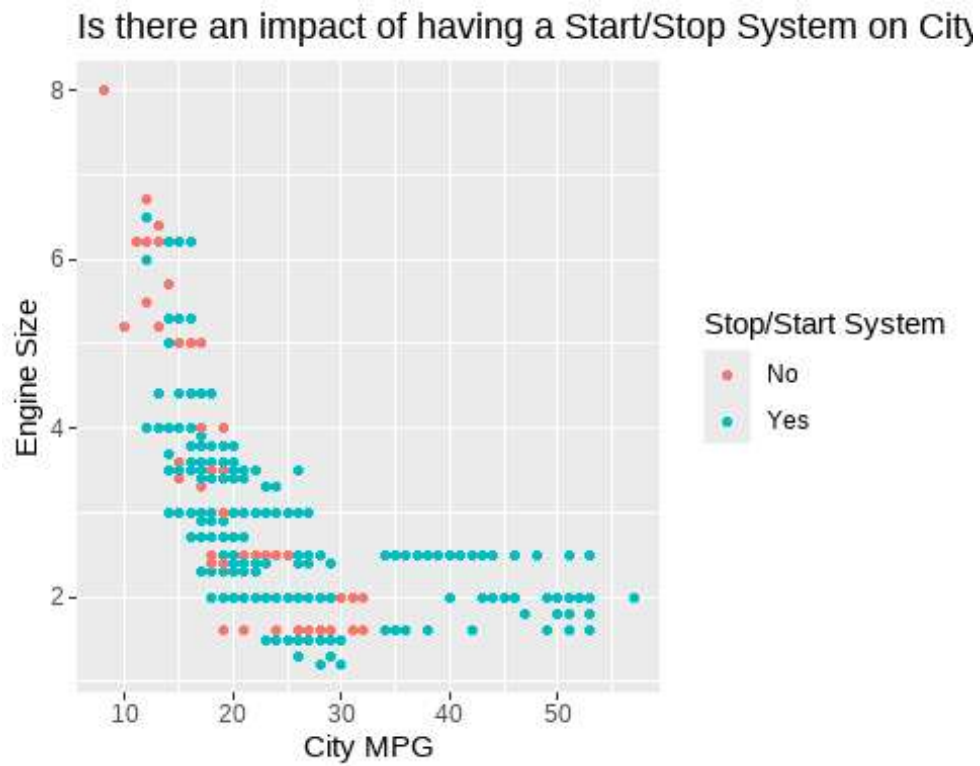
ggplot(efficiency, aes(y = reorder(division, avg_space_per_mpg), x =
  avg_space_per_mpg)) +
  geom_col() +
  labs(y = "Division",
    x = expression("Avg Space per MPG (ft\"^3\"/MPG)")) +
  theme_bw()
```



Q2: Is there a significant impact of the presence of Stop/Start system on City MPG (where stopping occurs more often) across different engine sizes?

```
ggplot(cars_fe, aes(x= city_fe, y= engine_displacement, color=
  stop_start_system_status)) +
  geom_point() +
```

```
labs(
  title = "Is there an impact of having a Start/Stop System on City MPG
across engine sizes?",
  x = "City MPG",
  y= "Engine Size",
  color = "Stop/Start System"
)
```



Follow-up Questions

New Questions Based Off Initial Investigation

- Q1: Is the fuel-cost advantage of Stop/Start systems consistent across vehicle classes, or concentrated in specific ones? (Extension of Q8)
- Q2: Do vehicle class and drive type together explain variation in combined fuel economy beyond what either explains alone? (Extension of Q1 and Q5)
- Q3: Does air aspiration type moderate the relationship between engine displacement and annual fuel cost? (Extension of Q4 and Q5)
- Q4: Is there a relationship between vehicle class and fuel-saving technologies (Stop/Start, turbocharging, and hybrid descriptors), and do those technology combinations predict better fuel economy outcomes? (Extension of Q6 and Q8)

Investigation of Follow-up Questions

GIVE WHAT 2 QUESTIONS YOU ATTEMPTED TO INVESTIGATE FURTHER IN COMPLETE SENTENCES

- Our group decided to investigate Q1 and Q3 in further detail.

SHOW AT LEAST 2 TABLES OR FIGURES BELOW THAT EXPLORE ANSWERS FOR THE QUESTIONS YOU ARE INVESTIGATING FURTHER.

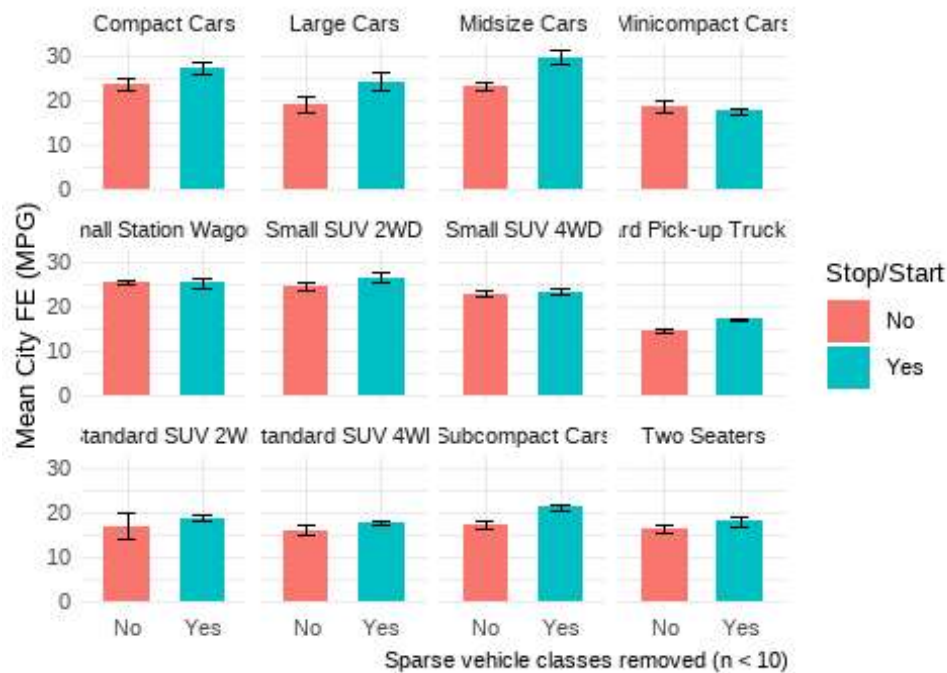
Q1: Is the fuel-cost advantage of Stop/Start systems consistent across vehicle classes, or concentrated in specific ones?

```
stop_start_by_class <- cars_fe |>
  select(
    carline_class_desc,
    stop_start_system_status,
    city_fe
  )
stop_start_by_class <- stop_start_by_class |>
  filter(!carline_class_desc %in% c(
    "Special Purpose Vehicle 2WD",
    "Special Purpose Vehicle, minivan 4WD",
    "Small Pick-up Trucks 2WD",
    "Special Purpose Vehicle cab chassis",
    "Midsize Station Wagons",
    "Special Purpose Vehicle, minivan 2WD",
    "Small Pick-up Trucks 4WD",
    "Standard Pick-up Trucks 2WD"))

stop_start_by_class |>
  filter(!is.na(stop_start_system_status)) |>
  ggplot(aes(x = stop_start_system_status,
             y = city_fe,
             fill = stop_start_system_status)) +
  stat_summary(fun = mean, geom = "bar", width = 0.6) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.25, linewidth
= 0.5) +
  facet_wrap(~ carline_class_desc, ncol = 4) +
  labs(
    title = "City FE by Stop/Start System Presence Across Vehicle Classes",
    subtitle = "Bars show mean city MPG, lines above show standard error ·
2025 EPA Fuel Economy Data",
    x = NULL,
    y = "Mean City FE (MPG)",
    fill = "Stop/Start",
    caption = "Sparse vehicle classes removed (n < 10)"
  ) +
  theme_minimal(base_size = 10)
```

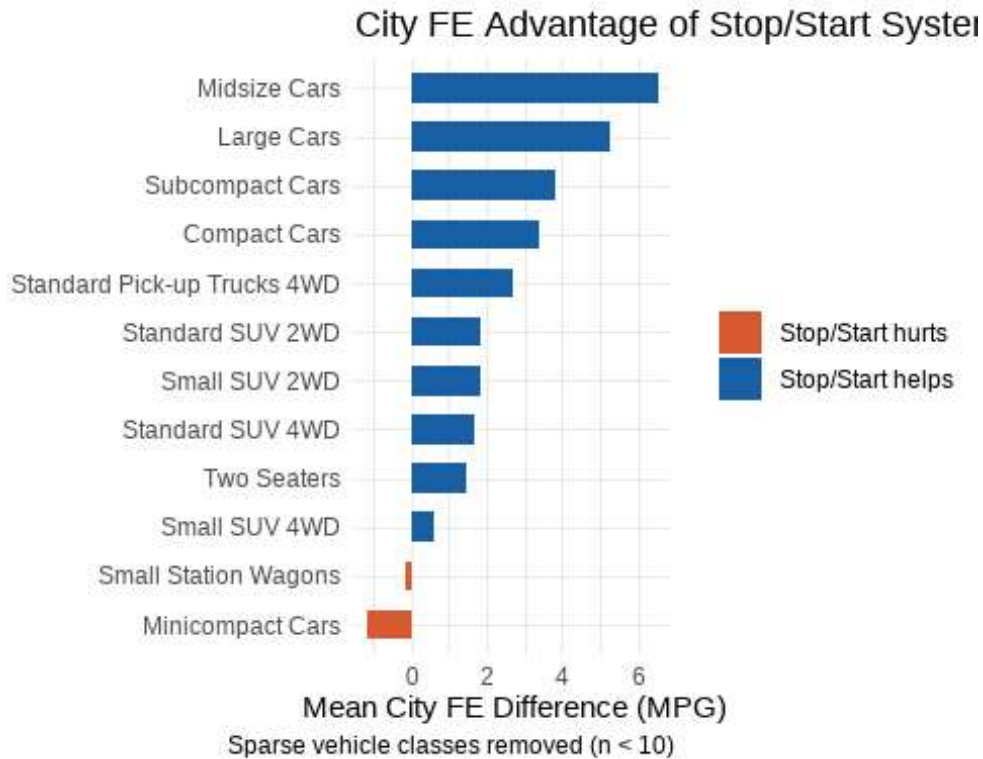

City FE by Stop/Start System Presence Across Vehicle Classes

Bars show mean city MPG, lines above show standard error - 2025 EPA Fuel Economy



```
stop_start_diff <- stop_start_by_class |>
  filter(!is.na(stop_start_system_status)) |>
  group_by(carline_class_desc, stop_start_system_status) |>
  summarise(mean_city_fe = mean(city_fe, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(names_from = stop_start_system_status,
              values_from = mean_city_fe) |>
  mutate(diff = Yes - No)

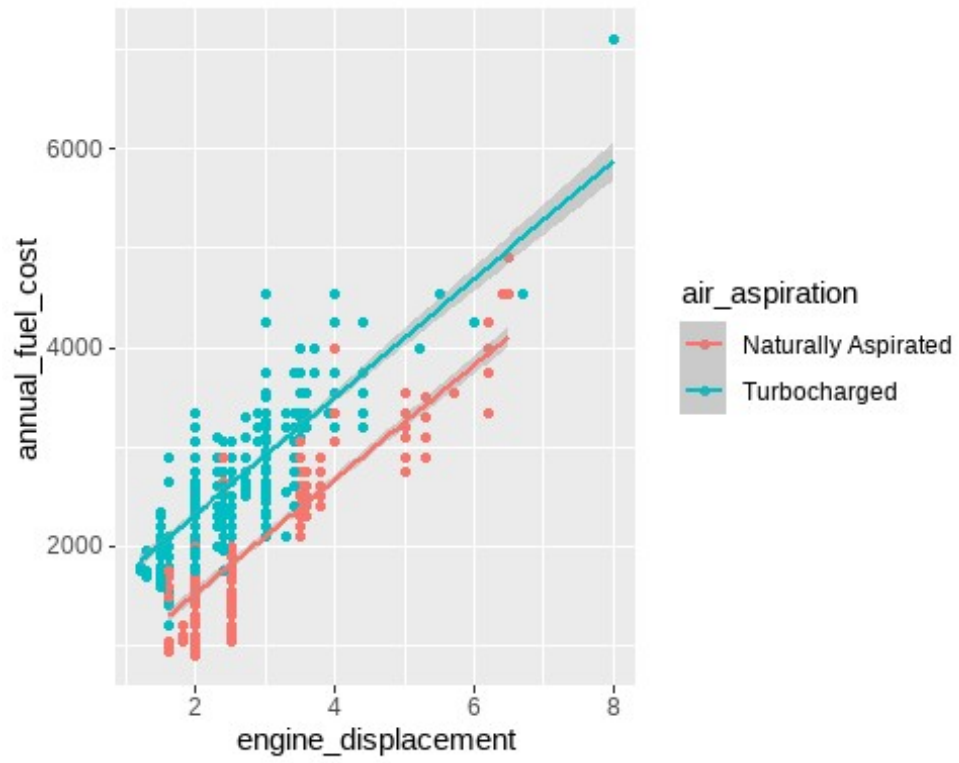
ggplot(stop_start_diff, aes(x = diff, y = reorder(carline_class_desc, diff),
  fill = diff > 0)) +
  geom_col(width = 0.6) +
  scale_fill_manual(values = c("TRUE" = "#185FA5", "FALSE" = "#D85A30"),
                    labels = c("TRUE" = "Stop/Start helps", "FALSE" =
"Stop/Start hurts"),
                    name = NULL) +
  labs(
    title = "City FE Advantage of Stop/Start System by Vehicle Class",
    x = "Mean City FE Difference (MPG)",
    y = NULL,
    caption = "Sparse vehicle classes removed (n < 10)"
  ) +
  theme_minimal()
```



Q3: Does air aspiration type moderate the relationship between engine displacement and annual fuel cost? (Extension of Q4 and Q5)

```
aspiration_data <- cars_fe |>
  select(
    engine_displacement,
    air_aspiration,
    annual_fuel_cost
  ) |>
  filter(
    air_aspiration %in% c("Turbocharged", "Naturally Aspirated")
  )

aspiration_data |>
  ggplot(aes(x = engine_displacement, y = annual_fuel_cost, color =
    air_aspiration)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1)
```



Summary

GIVE A 2 PARAGRAPH SUMMARY.

PARAGRAPH 1 SHOULD DESCRIBE WHAT YOU LEARNED ABOUT YOUR DATA FROM INVESTIGATING THE INITIAL QUESTIONS. DID YOU FIND ANYTHING UNUSUAL IN YOUR DATA? DID ANYTHING SURPRISE YOU? WHICH OF THE INITIAL QUESTIONS WERE HELPFUL IN LEADING YOU TO MORE QUESTIONS?

PARAGRAPH 2 SHOULD SUMMARIZE WHAT YOU LEARNED FROM INVESTIGATING THE FOLLOW-UP QUESTIONS. WHY ARE THESE FOLLOW-UP QUESTIONS INTERESTING FOR INVESTIGATION? DESCRIBE THE TABLES/FIGURES YOU USED TO EXPLORE ANSWERS TO THESE FOLLOW-UP QUESTIONS? WHAT DID YOU LEARN FROM THE TABLES/FIGURES REGARDING THE FOLLOW-UP QUESTIONS YOU PROPOSED?

One of the initial questions that we investigated was about the impact of Start/Stop systems across different engine sizes on city fuel efficiency. Our observations from this visualization were that cars with a start/stop system tend to have a much higher city MPG than cars without it across engine sizes with the difference being most noticeable in engines with smaller displacement. This led us to develop more questions about whether the results from this question are consistent across vehicle classes or concentrated in specific ones. One interesting thing we observed came from investigating the question about which factors could be used to predict the cost of fuel, the correlation matrix shows that the features most correlated with `annual_fuel_cost` are `engine_displacement` and `num_of_cylinders`, meaning that larger, more powerful engines will often increase the amount spent on fuel. Another question that led us to more questions was about the correlation between engine displacement and combined fuel efficiency when comparing different drive types (AWD vs FWD vs RWD). All of the drive types had a negative correlation with combined FE, but FWD seemed to have a slightly higher combined FE than other drive types.

Our two follow-up questions investigated whether the impact of Start/Stop systems on city MPG was consistent across vehicle classes or concentrated in specific ones. We created a set of visualizations which show the difference in mean and variability of cars with Start/Stop systems and those without, separated by vehicle classes. From this we found that Stop/Start systems help in most vehicle classes, with the highest advantage for midsize and large cars. It also hurts fuel efficiency in minicompact cars and small station wagons. We also investigated if air aspiration type moderates the relationship between engine displacement and annual fuel cost. We created a dot plot visualization that showed engine displacement and annual fuel cost on the axes, with the points that were plotted being separated by color based on the air aspiration type which allowed us to see the impact of air aspiration type on the relationship between engine displacement and fuel cost. The graph shows a clear split between naturally aspirated and turbocharged cars. Both regression lines are roughly parallel with the turbo line being higher meaning

turbocharged cars generally cost more than naturally aspirated as the engine displacement rises.